Highlights

- Weights sampled negatives by the posterior that an unlogged pair is positive.
- A fixed imputation-logit offset adds a gain without uniform training data.
- Beats every graph baseline in mean nDCG on Coat, Yahoo R3, KuaiRand, KuaiRec.
- Best overall on Yahoo R3 and KuaiRand; not above iALS or DR-JL on Coat or KuaiRec.
- Shuffled-weight controls and seed repeats show which gains are robust.